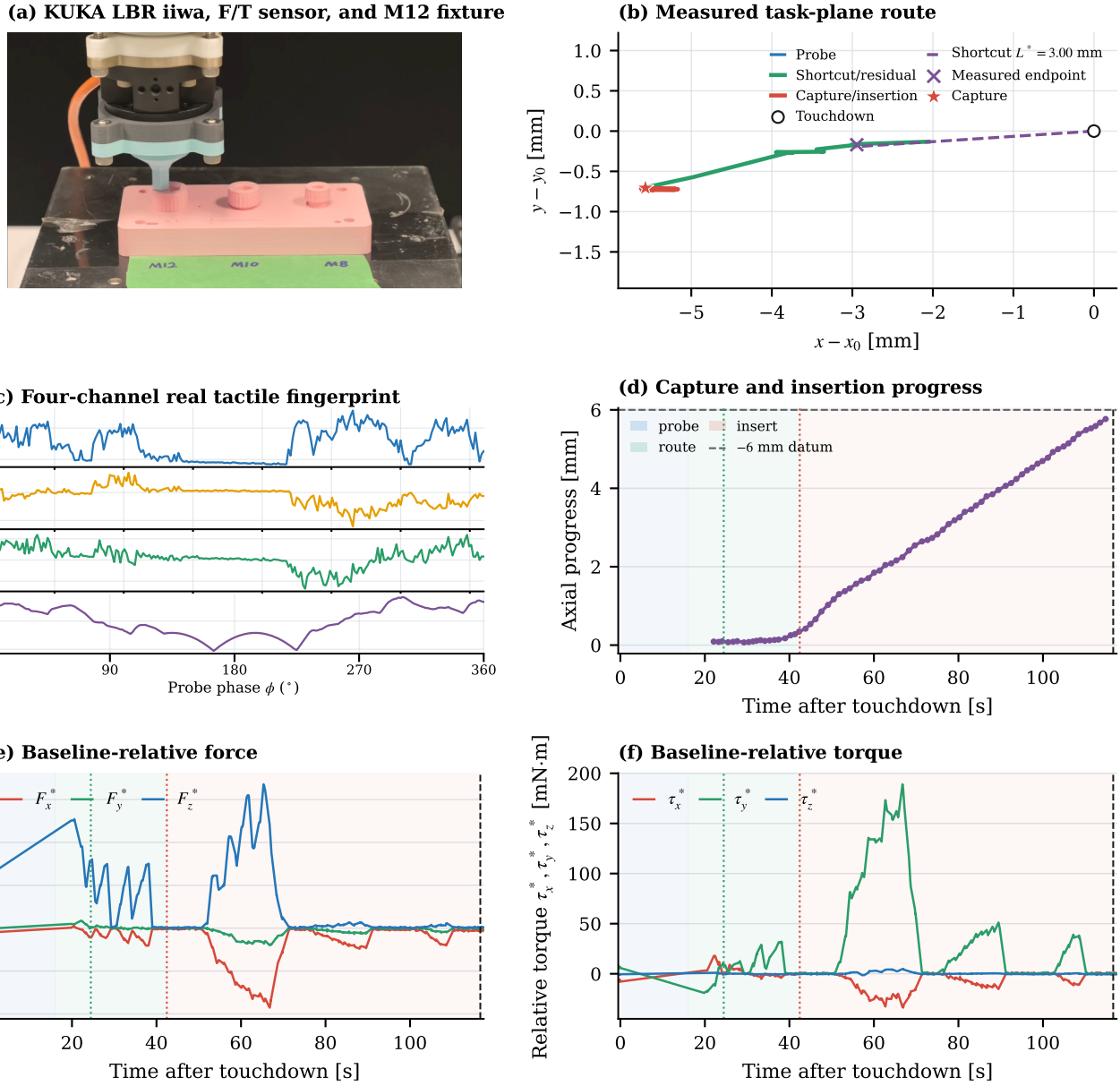

(a) KUKA LBR iiwa, F/T sensor, and M12 fixture
M12
M10
M8
(b) Measured task-plane route
Probe
Shortcut/residual
Capture/insertion
Touchdown
Shortcut $L^* = 3.00$ mm
Measured endpoint
Capture
$y - y_0$ [mm]
$x - x_0$ [mm]
(c) Four-channel real tactile fingerprint
F_z [N]
τ_x [mN·m]
τ_y [mN·m]
Δz [mm]
Probe phase ϕ (°)
(d) Capture and insertion progress
probe
route
insert
−6 mm datum
Axial progress [mm]
Time after touchdown [s]
(e) Baseline-relative force
F_x^*
F_y^*
F_z^*
Relative force F_x^*, F_y^*, F_z^* [N]
Time after touchdown [s]
(f) Baseline-relative torque
τ_x^*
τ_y^*
τ_z^*
Relative torque $\tau_x^*, \tau_y^*, \tau_z^*$ [mN·m]
Time after touchdown [s]